RACE
Regular Analysis Through Continuous Exercise

Differential Equations: RACE-01

Single Correct Question :

1. Find the order and degree of differential equation $\sin^{-1}\left(\frac{dy}{dx}\right)=x+y+3$

(A) order : 1 ; degree : not defined (B) order : 1 ; degree : 1
(C) order : 1 ; degree : 2 (D) None of these

2. Find the order and degree of differential equation $\left[\frac{d^2x}{dt^2}\right]^3+\left[\frac{dx}{dt}\right]^4-xt=0$

(A) order : 3 ; degree : 2 (B) order : 2 ; degree : 4
(C) order : 2 ; degree : 3 (D) order : 4 ; degree : 2

3. Eliminate the arbitrary constants and obtain the differential equation satisfied by $y = 2x + ce^x$

(A) $y' + y = 2(1 - x)$ (B) $y' - y = 2(1 - x)$
(C) $y' - y = 2(1 + x)$ (D) $y' + y = 2(1 + x)$

4. Eliminate the arbitrary constants and obtain the differential equation satisfied $y=\left(\frac{a}{x^2}\right)+bx$

(A) $x^2y" - 2xy' - 2y = 0$ (B) $x^2y" + 2xy' + 2y = 0$
(C) $x^2y" + 2xy' + 2y = 0$ (D) $x^2y" + 2xy' - 2y = 0$

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5. Eliminate the arbitrary constants and obtain the differential equation satisfied $y = ae^{2x} + be^{-2x} + c$

(A) $y''' = 4y'$ (B) $y' = 4y'$ (C) $y'' = 4y'$ (D) $y''' = 2y$

6. Obtain the differential equation of the family of circles $x^2 + y^2 + 2gx + 2fy + c = 0$; where g , f & c are arbitrary constants.

(A) $[1 + (y')^2].y''' - 3y'(y'')^2 = 0$ (B) $[1 + (y')^2].y''' + (y''+2y')(y'')^2 = 0$
(C) $[1 + (y')^2].y''' + 3y'(y'')^2 = 0$ (D) $[1 - (y')^2].y''' + 3y'(y'')^2 = 0$

7. Obtain the differential equation associated with the primitive, $y = c_1 e^{3x} + c_2 e^{2x} + c_3 e^x$, where c_1 , c_2 , c_3 are arbitrary constants.

(A) $\frac{d^3y}{dx^3}-6\frac{d^2y}{dx^2}+11\frac{dy}{dx}-6y=0$ (B) $\frac{d^3y}{dx^3}-6\frac{d^2y}{dx^2}-11\frac{dy}{dx}-6y=0$
(C) $\frac{d^3y}{dx^3}+6\frac{d^2y}{dx^2}+11\frac{dy}{dx}+6y=0$ (D) $\frac{d^3y}{dx^3}+6\frac{d^2y}{dx^2}-11\frac{dy}{dx}-6y=0$

8. Solution of differential equation $\sqrt{1+4x^2}dy = y^3xdx$ is

(A) $\frac{1}{2y^2}=\frac{1}{4}\sqrt{1+4x^2}+C$ (B) $\frac{1}{2y^2}=\frac{1}{4}\sqrt{1-4x^2}+C$
(C) $\frac{-1}{2y^2}=\frac{1}{4}\sqrt{1-2x^2}+C$ (D) $\frac{-1}{2y^2}=\frac{1}{4}\sqrt{1+4x^2}+C$

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9. Solution of differential equation $(1 - x^2)(1 - y)\,dx = xy(1 + y)\,dy$ is

(A) $\ell n(1+y)^2x = c - \frac{1}{2}y^2 + 2y + \frac{1}{2}x^2$ (B) $\ell n(1-y)^2x = c - \frac{1}{2}y^2 - 2y + \frac{1}{2}x^2$

(C) $\ell n(1-y)^2x = c + \frac{1}{2}y^2 + 2y + \frac{1}{2}x^2$ (D) $\ell n(1+y)^2x = c - \frac{1}{2}y^2 - 2y - \frac{1}{2}x^2$

10. Solution of differential equation $\frac{dy}{dx} + \frac{\sqrt{(x^2-1)(y^2-1)}}{xy} = 0$ is

(A) $\sqrt{x^2-1} - \sec^{-1}x + \sqrt{y^2-1} = c$ (B) $\sqrt{x^2-1} + \sec^{-1}x - \sqrt{y^2-1} = c$

(C) $\sqrt{x^2+1} - \sec^{-1}x + \sqrt{y^2+1} = c$ (D) $\sqrt{x^2-1} - \sec^{-1}x - \sqrt{y^2-1} = c$

Integer Type Questions :

11. Given $y(0) = 2000$ and $\frac{dy}{dx} = 32000 - 20y^2$, then find the value of $\lim_{x\to\infty} y(x)$

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Answer Key

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	A	C	B	D	A	A	A	D	B	A	40

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Differential Equations: RACE-02

Single Correct Questions :

1. Solution of the differential equation $\tan^2(x+y)dx - dy = 0$ is

(A) $2(x+y) = c + \sin 2(x+y)$ (B) $2(x-y) = c + \sin 2(x+y)$

(C) $2(x-y) = c + \sin 2(x-y)$ (D) $2(x+y) = c - \sin 2(x+y)$

2. Solution of the differential equation $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$ is

(A) $\tan\left(\frac{x+y}{2}\right) = ae^x - 1$ (B) $\tan\left(\frac{x-y}{2}\right) = ae^x - 1$

(C) $\tan\left(\frac{x+y}{2}\right) = ae^{-x} - 1$ (D) $\tan\left(\frac{x+y}{2}\right) = ae^x + 1$

3. Solution of the differential equation $\sin^{-1}\left(\frac{dy}{dx}\right) = x+y$ is

(A) $\tan(x+y) - \sec(x+y) = x + c$ (B) $\tan(x+y) + \sec(x+y) = x + c$

(C) $\tan(x+y) - \sec(x+y) + x = c$ (D) $\tan(x+y) + \sec(x+y) + x = c$

4. Solution of the differential equation $\frac{dy}{dx} = \frac{x-2y+5}{2x+y-1}$ is

(A) $x^2 + 4xy - y^2 - 10x + 2y = c'$ (B) $x^2 - 4xy + y^2 + 10x - 2y = c'$

(C) $x^2 - 4xy - y^2 + 10x + 2y = c'$ (D) $x^2 + 4xy + y^2 - 10x - 2y = c'$

5. Solution of the differential equation $\frac{dy}{dx} = \frac{2x-y+3}{x+2y+4}$ is

(A) $y^2 + x^2 - xy + 4y - 3x + c = 0$ (B) $y^2 - x^2 + xy - 4y + 3x + c = 0$

(C) $y^2 - x^2 + xy + 4y - 3x + c = 0$ (D) $y^2 + x^2 + xy - 4y + 3x + c = 0$

6. Solution of the differential equation $(y - xy^2)dx - (x + x^2y)\,dy = 0$ is

(A) $x = cye^{-xy}$ (B) $y = cxe^{xy}$ (C) $x^2 = cye^{xy}$ (D) $x = cye^{xy}$

7. Solution of the differential equation $ydx - xdy = xydy - x^2dx$ is

(A) $\sqrt{x^2 - y^2} = \sin^{-1}\left(\frac{y}{x}\right) + c$ (B) $\sqrt{x^2 + y^2} = \sin^{-1}\left(\frac{y}{x}\right) + c$

(C) $\sqrt{x^2 - y^2} = \sin^{-1}\left(\frac{x}{y}\right) + c$ (D) $\sqrt{x^2 + y^2} = -\sin^{-1}\left(\frac{y}{x}\right) + c$

8. Solution of the differential equation $xdx + y\,dy = x\,(xdy - ydx)$ is

(A) $(y-1)^2 = c(x^2 + y^2)$ (B) $(y+1)^2 = c(x^2 + y^2)$

(C) $(y+1)^2 = c(x^2 - y^2)$ (D) $(y-1)^2 = c(x^2 - y^2)$

Answer Key

Question	1	2	3	4	5	6	7	8
Answer	B	A	A	C	C	D	A	B

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Differential Equations: RACE-03

Single Correct Questions :

1. Solution of the differential equation $y' = \frac{3x-y}{x+y}$ is

(A) $(3x+y)(x-y)=c$ (B) $(3x+y)(x+y)=c$

(C) $(3x-y)(x-y)=c$ (D) $(x+3y)(x-y)=c$

2. Solution of the differential equation $(x - y\ell ny + y\ell nx)\,dx + x(\ell ny - \ell nx)\,dy = 0$ is

(A) $y\ell n\left(\frac{y}{x}\right)+y+x\ell nx+cx=0$ (B) $y\ell n\left(\frac{y}{x}\right)-y+x\ell nx+cx=0$

(C) $y\ell n\left(\frac{y}{x}\right)-y-x\ell nx+cx=0$ (D) $y\ell n\left(\frac{x}{y}\right)-y+x\ell nx+cx=0$

3. Solution of the differential equation $(3xy+y^2)dx + (x^2+xy)dy = 0, y(1)=1$ is

(A) $x^2y(2x-y)=3$ (B) $x^2y(x+2y)=3$

(C) $x^2y(2x+y)=3$ (D) $x^2y(2x+y)=0$

4. Solution of the differential equation $\frac{dy}{dx}=\frac{x+2y-5}{2x+y-4}$ is

(A) $x-y-3=C(x-y+1)^3$ (B) $x+y-3=C(x+y+1)^3$

(C) $x+y-5=C(x-y+1)^3$ (D) $x+y-3=C(x-y+1)^3$

5. Solution of the differential equation $\frac{dy}{dx}=\frac{x+y+1}{2x+2y+3}$ is

(A) $x+y+\frac{4}{3}=ce^{4(x-2y)}$ (B) $x+y+\frac{4}{3}=ce^{3(x+2y)}$

(C) $x+y+\frac{4}{3}=ce^{3(x-2y)}$ (D) $x-y+\frac{4}{3}=ce^{3(x-2y)}$

6. Solution of the differential equation $(x-y)dy = (x+y+1)dx$ is

(A) $\arctan\frac{2y+1}{2x+1}=\ell nc\sqrt{x^2-y^2+x+y+\frac{1}{2}}$

(B) $\arctan\frac{2y+1}{2x+1}=\ell nc\sqrt{x^2+y^2+x+y+\frac{1}{2}}$

(C) $\arctan\frac{2x+1}{2y+1}=\ell nc\sqrt{x^2+y^2+x+y+\frac{1}{2}}$

(D) $\arctan\frac{2y+1}{2x+1}=\ell nc\sqrt{x^2+y^2+x-y-\frac{1}{2}}$

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7. Solution of the differential equation $(x-a)\dfrac{dy}{dx}+3y=12(x-a)^3, x>a>0$ is

(A) $y=2(x-a)^3+\dfrac{c}{(x+a)^3}$ (B) $y=2(x+a)^3+\dfrac{c}{(x-a)^3}$

(C) $y=2(x-a)^3+\dfrac{c}{(x-a)^3}$ (D) $y=3(x-a)^3+\dfrac{c}{(x-a)^3}$

8. Solution of the differential equation $y\,\ell n\,y\,dx+(x-\ell n\,y)\,dy=0$ is

(A) $2x\,\ell n\,y=\ell n^2\,x+C$ (B) $2x\,\ell n\,y=\ell n^2\,y+C$

(C) $2y\,\ell n\,y=\ell n^2\,y+C$ (D) $4x\,\ell n\,y=\ell n^2\,y+C$

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Answer Key

Question	1	2	3	4	5	6	7	8
Answer	A	B	C	D	C	B	C	B

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Differential Equations: RACE-04

Single Correct Questions :

1. $x(x-1)\frac{dy}{dx}-(x-2)y=x^3(2x-1)$

(A) $y(x-1)=x^2(x^2-x+c)$ (B) $y(x+1)=x^2(x^2-x+c)$

(C) $y(x-1)=x^2(x^2+x+c)$ (D) $y(x+1)=x^2(x^2+x+c)$

2. $xdy-\{y+xy^3(1+\ell nx)\}dx=0$

(A) $\frac{x^2}{y^2}=\frac{2}{3}x^3\left(\frac{2}{3}+\ell nx\right)+c$ (B) $\frac{x^2}{y^2}=-\frac{2}{3}x^3\left(\frac{2}{3}+\ell nx\right)+c$

(C) $\frac{x^2}{y^2}=-\frac{2}{3}x^3\left(\frac{1}{3}+\ell nx\right)+c$ (D) $\frac{x^2}{y^2}=-\frac{3}{2}x^3\left(\frac{2}{3}+\ell nx\right)+c$

3. $x^2y-x^3\frac{dy}{dx}=y^4\cos x$

(A) $x^3y^{-3}=3\cos x+c$ (B) $x^{-3}y^3=3\sin x+c$

(C) $x^3y^3=3\sin x+c$ (D) $x^3y^{-3}=3\sin x+c$

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4. $x(x^2+1)\frac{dy}{dx}=y(1-x^2)+x^3.\ell nx$

(A) $4(x^2+1)y-x^3(1-2\,\ell nx)=cx$ (B) $4(x^2+1)y+x^3(1+2\,\ell nx)=cx$

(C) $4(x^2+1)y+x^3(1-2\,\ell nx)=cx$ (D) $(x^2+1)y+x^3(1-2\,\ell nx)=cx$

5. $\left(\frac{dy}{dx}\right)^2-(x+y)\frac{dy}{dx}+xy=0$

(A) $y=ce^x\ ;\ y=c+\frac{x^2}{2}$ (B) $y=ce^x\ ;\ y=c-\frac{x^2}{2}$

(C) $x=ce^y\ ;\ y=c+\frac{x^2}{2}$ (D) $y=ce^x\ ;\ y=c+\frac{x^2}{4}$

6. $\frac{dy}{dx}=e^{x-y}(e^x-e^y)$

(A) $e^y=c.\exp(-e^x)+e^x-1$ (B) $e^y=c.\exp(-e^x)+e^x+1$

(C) $e^y=c.\exp(e^x)+e^x-1$ (D) $e^y=c.\exp(e^x)-e^x-1$

7. $x\,dx+y\,dy+4y^3(x^2+y^2)dy=0.$

(A) $\ell n(x^2+y^2)+y^4=C$ (B) $1/3\ \ell n(x^2+y^2)+y^4=C$

(C) $1/2\ \ell n(x^2+y^2)+y^4=C$ (D) $1/2\ \ell n(x^2-y^2)+y^4=C$

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Integer Type Questions :

8. Let f be a differentiable function such that $x^2 f(x) - x = 4\int_0^x t\, f(t)dt$, $f(1)=\frac{2}{3}$.

Then $18\, f(3) =$ _____

9. If $y = y(x)$ is the solution of the differential equation $\frac{dy}{dx}+\frac{4x}{(x^2-1)}y=\frac{x+2}{(x^2-1)^{\frac{5}{2}}}$, $x > 1$ such that $y(2)=\frac{2}{9}\log_e\left(2+\sqrt{3}\right)$ and $y\left(\sqrt{2}\right) = \alpha\log_e\left(\sqrt{\alpha}+\beta\right)+\beta-\sqrt{\gamma}, \alpha,\beta,\gamma\in N$, then $\alpha\beta\gamma$ is equal to ___.

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Answer Key

Question	1	2	3	4	5	6	7	8	9
Answer	A	B	D	C	A	A	C	160	6

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